HERIOT-WATT UNIVERSITY

F79MB Statistical Models B

Assessment Project 1 – 2025: Mark scheme

1. Numerical summary of the medical costs billed by private hospitals across various regions of the United States (in USD) for the year 2024:

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Min.   1st Qu.  Median    Mean    3rd Qu.   Max.
1137    3947     9229     13412    18158    51195
```

Sample standard deviation is `$12225.63`.

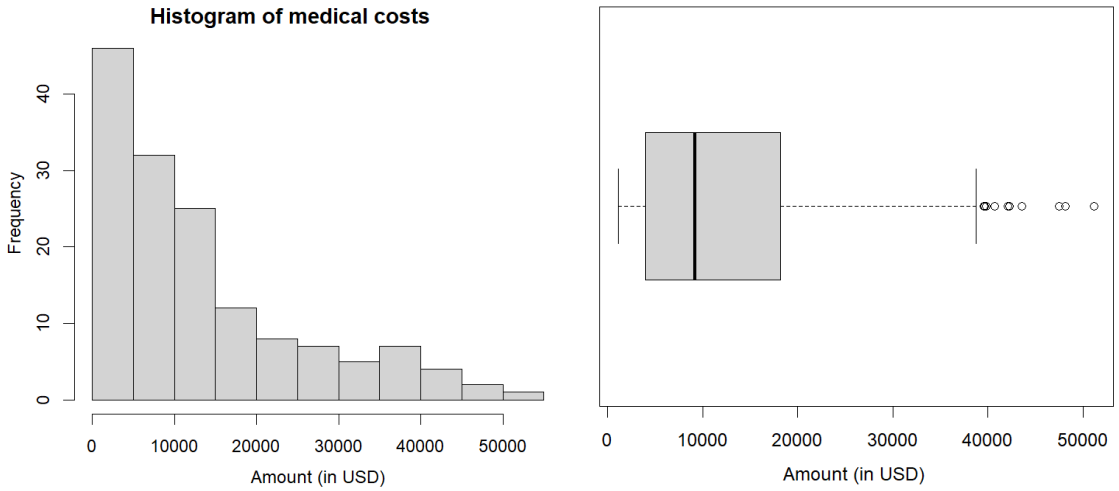


Figure 1: Histogram and boxplot of medical costs (in USD) for private hospitals across various regions of the United States in 2024, based on a sample of 149 patients.

Comment: From the histogram (Figure 1), the distribution appears unimodal and highly right-skewed, with the highest medical cost at $51,195. Numerical summaries also indicate mean is considerably larger than the median. The minimum travel expenses are $1,137 and the standard deviation is $12,225. The boxplot shows the presence of some outliers at the upper end of the distribution.

2.

(i) Using the MLE estimate for the exponential distribution where $\hat{\lambda} = \frac{1}{\bar{X}} = 7.456\times 10^{-5}$, the original data seems not closely follow an exponential distribution. Although the QQ plot (Figure 2(i)) shows that most points on the lower tail fall along the line $Y = X$, some points in the middle deviate slightly above the upper part of the line, and approximately 6 to 7 points at the upper tail deviate below the line.

(ii) The QQ plot for the log-transformed data (Figure 2(ii)) appears somewhat closer to a normal distribution with a mean of 9 and variance of 1. However, a few points on the lower tail lie slightly above the straight line, and a few points on the upper tail lie below the line ($Y = X$). The histogram and boxplot in Figure 3 further support that the data does not deviate significantly from the given normal distribution for the log-transformed data. Both plots show minimal skewness in the log-transformed data.

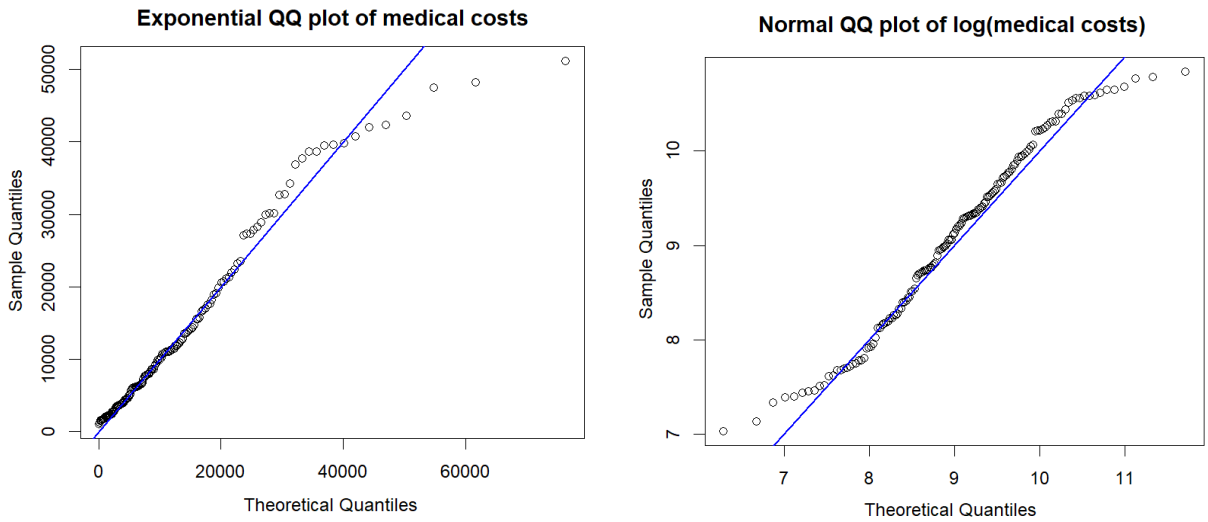


Figure 2: (i) Exponential QQ plot for original data (top left); (ii) Normal QQ plot for log-transformed data.

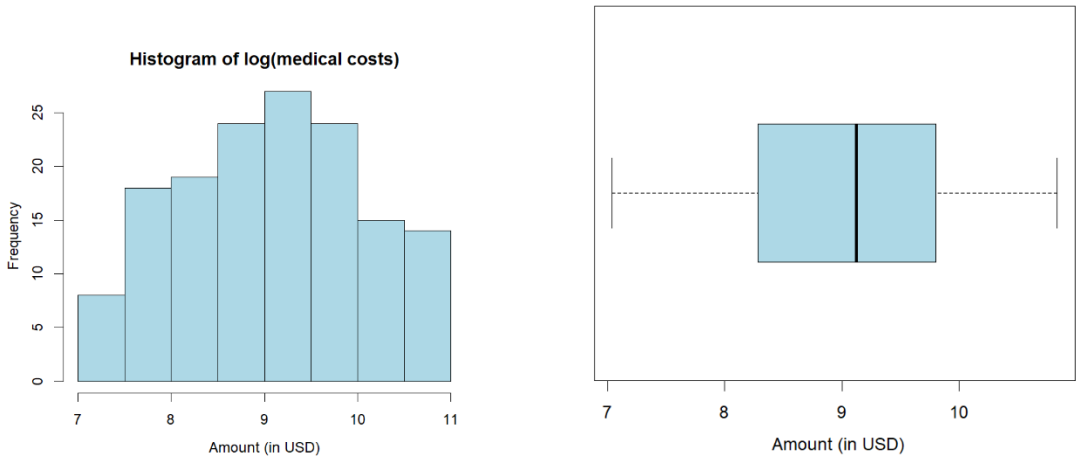


Figure 3: Histogram and boxplot for log-transformed data.

3.(i) Perform a χ^2 goodness-of-fit

H_0: The (original) medical costs data come from an Exponential distribution.
H_1: These (original) medical costs data do not come from an Exponential distribution.

Let the original data be $x_1, x_2, \cdots, x_n$. The null hypothesis does not specify the parameters $\hat{\lambda}$ of an exponential distribution, so we estimate from the data using the typical MLE estimator as calculated in 2(i), $\hat{\lambda} = \frac{n}{\sum_{i=1}^{n} x_i} = 7.456 \times 10^{-5}$. A χ^2 test with 9 equal-probability cells, will give all expected frequencies $e_i = \frac{149}{9} = 16.5556$, $i = 1, 2, \ldots, 9$. The equal-probability intervals (computed in *R*), together with corresponding observed frequencies, are:

Interval (costs)	Observed frequency, f_i
$[0, 1.58 \times 10^3)$	3
$[1.58 \times 10^3\ , 3.37 \times 10^3)$	24
$[3.37 \times 10^3, 5.44 \times 10^3)$	21
$[5.44 \times 10^3,\ 7.88 \times 10^3)$	18
$[7.88 \times 10^3,\ 1.09 \times 10^4)$	16
$[1.09 \times 10^4,\ 1.47 \times 10^4)$	21
$[1.47 \times 10^4,\ 2.02 \times 10^4)$	12
$[\ 2.02 \times 10^4,\ 2.95 \times 10^4)$	14
$[2.95 \times 10^4, \infty)$	20

The test statistic value is $\chi^2 = \sum_{i=1}^{9} \frac{(f_i - e_i)^2}{e_i} = 19.34228$. With 9 cells and an estimated parameter $\hat{\lambda}$, the test statistic χ^2 distribution has *df* = 9 –1–1 = 7, so the *p*-value = $P(\chi_7^2 > 19.34228) = 0.00718$. There is evidence to reject H_0. Overall, it seems that the original data may not be well modelled by an Exponential distribution. It is worth noting that there is some evidence, as shown in the QQ plot (Figure 2(i)), that the data deviates from the Exponential distribution. While the bulk of the data fits well, particularly in the lower tail, there are still deviations observed.

3. (ii) Perform a χ^2goodness-of-fit

H_0: The log-transformed data come from a Normal distribution with a mean of 9 and variance of 1.

H_1: These log-transformed data come do not come from the given Normal distribution.

Similarly, a χ^2 test with 9 equal-probability cells, will give all expected frequencies $e_i = 16.5556$, $i = 1, 2, \ldots, 9$. The equal-probability intervals (computed in *R*), together with corresponding observed frequencies, are:

Interval (log(costs))	Interval (costs)	Observed frequency, f_i
$(-\infty,\ 7.78)$	$(0, 2.392 \times 10^3)$	19
$[7.78, 8.24)$	$[2.392 \times 10^3, 3.790 \times 10^3)$	16
$[8.24, 8.57)$	$[3.790 \times 10^3, 5.271 \times 10^3)$	13
$[8.57, 8.86)$	$[5.271 \times 10^3,\ 7.044 \times 10^3)$	14
$[8.86,\ 9.14)$	$[7.044 \times 10^3,\ 9{,}321 \times 10^3)$	13
$[9.14,\ 9.43)$	$[9.321 \times 10^3,\ 1.246 \times 10^4)$	19
$[9.43,\ 9.76)$	$[1.246 \times 10^4,\ 1.733 \times 10^4)$	15
$[\ 9.76,\ 10.2)$	$[\ 1.733 \times 10^4,\ 2.690 \times 10^4)$	17
$[10.2, \infty)$	$[2.690 \times 10^4, \infty)$	23

The test statistic value is $\chi^2 = \sum_{i=1}^{9} \frac{(f_i - e_i)^2}{e_i} = 5.28859$. Since all parameters of normal distribution are specified, the test statistic χ^2 distribution has *df* = 9 – 1 = 8, so the *p*-value = $P(\chi_8^2 > 5.328859) = 0.721918$. Thus, there is no evidence to reject H_0.

We can conclude that the log-transformed data can be modelled by a Normal distribution with a mean of 9 and variance of 1. However, as seen from the QQ plot (Figure 2(ii)), some points in the lower and upper parts of the plot do not fit the line very well.

Overall comment: Based on the Chi-squared goodness-of-fit test, the exponential distribution does not fit the medical costs data very well. This conclusion is further supported by the QQ plot in Figure 2(i), highlights deviations from the exponential distribution, particularly in the upper tail.

On the other hand, the log-transformed data appears to be adequately modelled by a normal distribution with a mean of 9 and a variance of 1, as indicated by the QQ plot, histogram, boxplot (Figures 2(ii) and 3), and the Chi-squared goodness-of-fit test. However, slight deviations are observed in the lower and upper tails of the QQ plot, and the histogram does not perfectly resemble a normal distribution.

4. If the parameters of the hypothesized distribution are known beforehand, the KS test can be applied. This makes the test valid for comparing the empirical distribution function (EDF) of the data with the specified cumulative distribution function (CDF).

Since the parameter λ for the exponential distribution is not specified in (i), we cannot apply the KS test. However, the parameters $\mu = 9$ and $\sigma = 1$ for the normal distribution are specified in (ii); therefore, the KS test is applicable in this case.

Carrying out the test in R, the value of the test statistic is computed as D = 0.076052, with a *p*-value of 0.3548. Thus, the log-transformed data appears reasonably well fit to the proposed Normal (9, 1) distribution. The result is consistent with the chi-squared test in part (3).

Comments for reduction of sample size: The reduction in sample size from 149 to 30 data points significantly impacts the results of the Kolmogorov-Smirnov (KS) test and the chi-squared goodness-of-fit test.

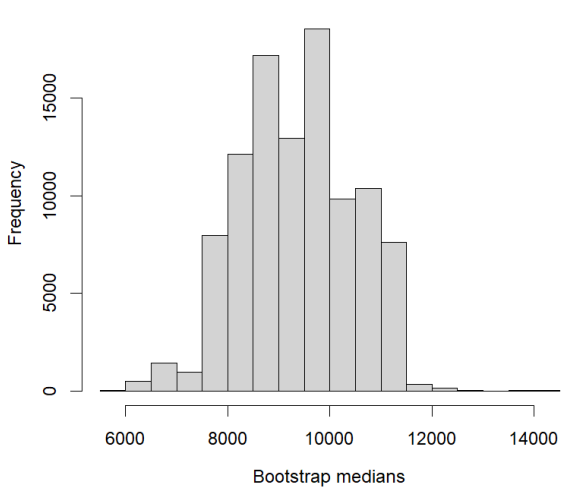
The KS test remains reliable with smaller sample sizes as it does not rely on binning. It works directly with the empirical distribution function (EDF) and compares it to the cumulative distribution function (CDF) of the hypothesized distribution. However, with fewer data points, the power of the KS test decreases, meaning it may not detect small deviations between the sample and the hypothesized distribution.

The reliability of the chi-squared goodness-of-fit test is significantly affected by the reduced sample size. The test requires sufficient expected frequencies (at least 5) in each bin to produce valid results. With only 30 data points, it becomes challenging to meet this criterion unless the number of bins is reduced. Reducing the number of bins can lead to a loss of granularity and sensitivity, potentially masking important differences between the observed and expected distributions. In extreme cases, the chi-squared test may not even be applicable.

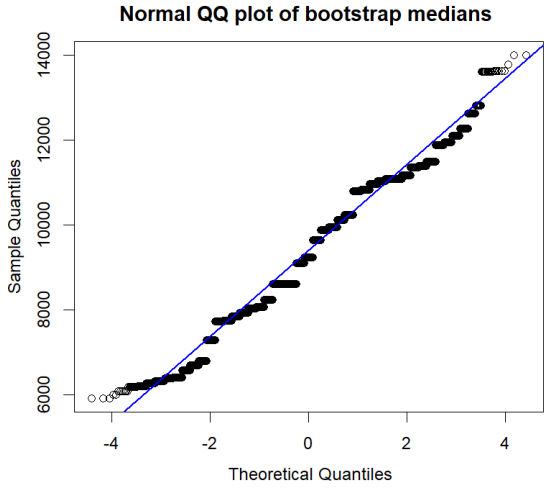
The KS test is more reliable for small samples, as it does not lose information through binning. The chi-squared test becomes unreliable for small samples unless careful adjustments are made, such as combining bins to ensure sufficient expected frequencies. In general, the KS test is better suited for smaller datasets, whereas the chi-squared test is more appropriate for larger datasets with well-defined bins.

5. The empirical (bootstrap) sampling distribution of the sample median, $\hat{m}$, is shown in Figure 4. The histogram and the QQ plot show a slightly positively skewed distribution. It is consistent with the mean is slightly larger than the median.

Bootstrap 95% CI for the population median medical costs is (7281.51, 11163.57).



(i)



(ii)

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   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   5920    8601    9229    9328   10115   14001
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(iii)

Figure 4: Distribution of bootstrap medians: (i) Histogram; (ii) Normal QQ plot (iii) Numerical summary

6. Under the asymptotic theory, we should have $\hat{m} \sim N(m, \left(\frac{1}{2f(m)\sqrt{n}}\right)^2)$, where $f(\hat{m})$ is the density of $N(9,1)$ distribution and $\hat{m}$ is the log-transformed sample median. We have $\hat{m}$=9.130089 and sample size n =149. From R, we obtain $f(\hat{m})$ = 0.3955808 and the estimate of the standard error of median is approximately given by $se(\hat{m}) = \frac{1}{2f(\hat{m})\sqrt{n}} = 0.103548$.

Then a 95% confidence interval (CI) of the log (population median medical costs), m, is

$$\hat{m} \pm 1.959964 \times se(\hat{m})$$

. $$(8.927,\ 9.333)$$

Taking the exponent of the CI above, the 95% CI of the population median medical costs is $e^{\hat{m} \pm 1.959964 \times se(\hat{m})}$, i.e.,

$$(7533.681,\ 11305.455).$$

The 95% CIs obtained in questions 5 and 6, using the non-parametric bootstrap and based on asymptotic theory, are slightly different but are considered to be not very far from each other. This is because the sampling distribution of the bootstrap medians (Figure 4(i)) is only slightly skewed to the right, and not too far from a normal distribution. With a sample size of n = 149 (quite large), the asymptotic theory is expected to perform well. Additionally, as shown in the histogram and boxplot in Figure 3, the log-transformed data do not raise concerns regarding normality assumption. The asymptotic theory CI is also narrower than the bootstrap CI. Of the two confidence intervals,

the one based on asymptotic theory is simpler and computationally more efficient, making it the preferred choice in this case.

7. We use a parametric bootstrap to perform a hypothesis test:

H_0: The population median medical costs is \$8,500 or more
H_1: The population median medical costs is less than \$8,500
[or equivalently, H_0: $Median \geq$ 8,500 against H_1: $Median <$ 8,500]

From the CDF of an exponential distribution with parameter λ, we have $F_X(x) = 1 - e^{-\lambda x}$. So, under H_0, where median $m_0 = 8500$, we can find $\lambda = \frac{\log(2)}{8500} = 8.15467 \times 10^{-5}$.

Figure 5 shows the sampling distribution of the sample median under the H_0 that data come from the Exponential (λ) with sample median is \$9,228.847. We find p-value = 0.7609. There is no evidence that the population median medical costs is less than \$8,500.

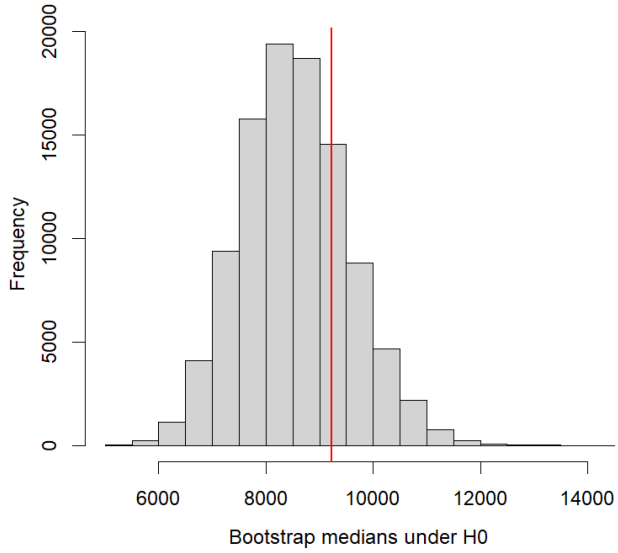


Figure 5: Sampling distribution of bootstrap medians of sample from Exponential (8.15467×10^{-5}) distribution, with red line showing sample median.

8. In conclusion, the medical costs (in USD) for private hospitals across various regions of the United States in 2024, based on a sample of 149 dataset exhibits a very positively skewed distribution, with a mean of \$13,412, a median of \$9,229, a minimum of \$1137, a maximum of \$51,195 and a standard deviation of \$12,225.63.

From the QQ plots, the chi-squared goodness-of-fit tests and the KS test, the log-transformed data appears reasonably modelled using a $N(9, 1)$ distribution. Note that the KS test can be applied to part 3(ii) where the parameters are specified (normal distribution), but it cannot be applied to part 3(i) with unspecified parameters (exponential distribution).

When we perform 95% confidence intervals for the population median using both the non-parametric and asymptotic theory methods, the intervals are not very far from each other. Of the two CIs, the one based on asymptotic theory is preferred because it is easier to compute, provides a more efficient and theoretically justified method for estimating the population median, and yields a narrower CI. Additionally, the sample size $n = 149$ is large, justifying the use of asymptotic theory for finding the confidence interval for the population median.

When performing the parametric bootstrap hypothesis test under the assumption that the original data follows an exponential distribution, there is no evidence to suggest that the population median of medical costs is less than $8,500.

Rubric Presentation

Overall presentation: 8 marks. Follow the rubric in Appendix B. Note that this rubric is written with the intention that 8 marks will only be awarded to an exceptionally good piece of work; I expect most students to get 6 marks; I expect a reasonable number of students to get 4 marks; and I hope only a very few students will get 0 or 2 marks.